

Pratheep Senthilkumar

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OBJECTIVE

Computer Science undergraduate specializing in AI & ML, with a strong interest in full-stack web development and applied machine learning. Motivated to build intelligent, end-to-end software systems that solve real-world problems. Passionate about continuously learning new technologies and applying strong problem-solving skills to create impactful solutions.

EDUCATION

B.Tech Computer Science and Engineering (AI & ML) 2023 – Present

Vellore Institute of Technology, Chennai

CGPA: 9.10

Class XII (CBSE) 2022 – 2023

Akshara Vidyalaya, Nellore

Percentage: 96.6

SKILLS

- **Programming Languages:** C, C++, Java, Python, JavaScript, SQL
- **Web Technologies:** HTML, CSS, React
- **Core Concepts:** Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, DSA, Machine Learning, Deep Learning
- **Tools & Methodologies:** scikit-learn, DevOps Fundamentals, Agile Methodology

PROJECTS

Brain Tumor MRI Classification using Hybrid Ensemble Learning

- Developed a deep learning-based brain tumor MRI classification system using CNN, InceptionV3, ConvNeXt, and Random Forest models.
- Implemented ensemble learning techniques to improve classification accuracy and model performance on MRI datasets.
- Built a Gradio-based interface for real-time tumor prediction and confidence score visualization using TensorFlow/Keras.

Weather-Wardrobe – Weather Recommendation REST API with CI/CD

- Developed a Python REST API for weather-based outfit recommendations using the OpenWeather API.
- Automated build and deployment workflows using GitHub Actions, Docker, and Ansible.
- Configured monitoring and performance visualization using Graphite and Grafana dashboards.

PUBLICATION

- Co-authored the IEEE SPERT 2025 paper, “**Efficient Deployment of Machine Learning Models for Tidal Current Prediction**”, focusing on tidal current and power forecasting using Random Forest Regressor (RFR) and GRU models.

CERTIFICATES

- **DevOps Fundamentals** — IBM Career Education Program
- **DevOps, Agile & Design Thinking** — IBM Career Education Program